Effective Eigenvalue Moduli on Compact Smoothness Classes

Optimal Hölder modulus, a gapped theorem, and sharp diagonal Sobolev rates

Candidate substantial partial result for Problem 1.15 in arXiv:2603.23056

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Status: candidate substantial partial result, likely valid. We give an optimal effective modulus into every $C^{0,\alpha}$ space, a locally Lipschitz $W^{1,q}$ modulus on uniformly gapped families, and a power-sharp $W^{1,q}$ modulus on the compact diagonal subclass of the source's $C^{1,\beta}$ ball. The general no-gap, nondiagonal $W^{1,q}$ modulus remains open.

1 Source problem and scope

For a Hermitian $d \times d$ matrix A, let

$$\mathcal{E}(A) = (\lambda_1(A), \dots, \lambda_d(A)), \qquad \lambda_1(A) \le \cdots \le \lambda_d(A),$$

and let $\mathcal{E}$ also denote the superposition map on matrix-valued functions. The source proves continuity of $\mathcal{E}$ in finite Sobolev topologies and $C^{0,\alpha}$ for $\alpha < 1$, but not uniform continuity on the entire Lipschitz space. It then singles out the relatively compact class

$$K_C^\beta = \{A \in C^{1,\beta}(\overline{U}, \mathrm{Herm}(d)) : \|A\|_{C^{1,\beta}} \le C\} \tag{1}$$

and poses the following problem on source PDF p. 9 [1].

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where $r = \frac{\alpha}{1-\alpha}$.

Even tough the maps in (2) and (3) are not uniformly continuous, it is very desirable to find effective moduli of continuity for their restrictions to interesting (for applications) compact subspaces of $C^{0,1}(\overline{U}, \mathrm{Herm}(d))$. For instance, for given $C > 0$ and $\beta \in (0, 1]$, the set

$$K := \left\{A \in C^{1,\beta}(\overline{U}, \mathrm{Herm}(d)) : \|A\|_{C^{1,\beta}(\overline{U}, M_d(\mathbb{C}))} \le C\right\}, \tag{1.10}$$

is a relatively compact subset of $C^{0,1}(\overline{U}, \mathrm{Herm}(d))$, by the Arzelà–Ascoli theorem. Thus, $\mathcal{E}|_K : K \to C_q^{0,1}(\overline{U}, \mathbb{R}^d)$, for $1 \le q < \infty$, and $\mathcal{E}|_K : K \to C^{0,\alpha}(\overline{U}, \mathbb{R}^d)$, for $0 < \alpha < 1$, are uniformly continuous, by Theorem 1.5 and Corollary 1.6.

Problem 1.15. *Find effective moduli of continuity for the restrictions of the characteristic map $\mathcal{E}$ to interesting compact subspaces.*

For instance, using the sequence A_n in (1.9) we will see in Example 7.4 that, for $1 \le q < \infty$ and $q^{-1} < \alpha \le 1$, the map $\mathcal{E}|_K : K \to C_q^{0,1}([0,1], \mathbb{R}^d)$ is not α-Hölder continuous.

Remark 1.16. Due to Rellich [Rel37], a real analytic curve of Hermitian matrices

Exact transcription of Problem 1.15. "Find effective moduli of continuity for the restrictions of the characteristic map $\mathcal{E}$ to interesting compact subspaces."

The problem is deliberately broad. We completely settle its $C^{0,\alpha}$ part on every Lipschitz-bounded class, give a complete $W^{1,q}$ answer on the compact diagonal subclass of (1), and give a useful gapped nondiagonal answer. We do not claim a general no-gap $W^{1,q}$ estimate.

Proof Intuition

There are two distinct sources of instability. For $C^{0,\alpha}$, no spectral information beyond Hoffman–Wielandt is needed: the difference of two eigenvalue maps is uniformly small and uniformly Lipschitz, and the elementary interpolation between those two facts gives the exact power $1 - \alpha$.

Derivatives are subtler because an eigenvalue crossing changes the sorting permutation. In the diagonal case every such change is a swap between two scalar functions whose difference is small. A one-dimensional coarea argument controls the derivative of a $C^{1,\beta}$ function on its thin sublevel set; a sorting network then adds the finitely many pairwise costs. Two examples expose the two competing scales: moving one crossing produces the exponent $1/q$, while small-amplitude high-frequency crossings produce $\beta/(1+\beta)$. Away from crossings, a uniform spectral gap makes the rank-one spectral projections Lipschitz, yielding a linear modulus.

2 Optimal modulus into $C^{0,\alpha}$

We use Frobenius norm for matrices and Euclidean norm for eigenvalue vectors. For a vector-valued function h on $\overline{U}$, write

$$\|h\|_{C^{0,\alpha}} = \|h\|_\infty + [h]_\alpha, \qquad [h]_\alpha = \sup_{x\neq y} \frac{\|h(x)-h(y)\|}{|x-y|^\alpha}.$$

Lemma 1 (Sup–Lipschitz interpolation)**.** *If h is bounded and Lipschitz and $0<\alpha<1$, then*

$$[h]_\alpha \le (2\|h\|_\infty)^{1-\alpha}\,\mathrm{Lip}(h)^\alpha. \tag{2}$$

Proof. For $r=|x-y|$, $\|h(x)-h(y)\| \le \min\{2\|h\|_\infty, \mathrm{Lip}(h)r\}$. The first bound is better above the intersection scale and the second below it; division by r^α gives (2). ☐

Theorem 2 (Optimal general Hölder modulus)**.** *Let $\mathcal{K} \subset C^{0,1}(\overline{U}, \mathrm{Herm}(d))$ satisfy $\mathrm{Lip}(A)\le C$ for all $A\in\mathcal{K}$. If $\delta = \|A-B\|_{C^{0,1}} \le 1$, then, for every $0<\alpha<1$,*

$$\|\mathcal{E}(A)-\mathcal{E}(B)\|_{C^{0,\alpha}} \le \delta + 2C^\alpha\delta^{1-\alpha}. \tag{3}$$

The power $1-\alpha$ cannot be replaced, even on one fixed bounded $C^{1,\beta}$ ball of real symmetric 2×2 curves.

Proof. Hoffman–Wielandt gives $\|\mathcal{E}(A)-\mathcal{E}(B)\|_\infty \le \|A-B\|_\infty \le \delta$ and $\mathrm{Lip}(\mathcal{E}(A)-\mathcal{E}(B)) \le \mathrm{Lip}(\mathcal{E}(A)) + \mathrm{Lip}(\mathcal{E}(B)) \le 2C$. Lemma 1 yields (3).

For sharpness, on an interval containing $[0,t]$, put

$$A_t(x) = \begin{pmatrix} x & 0 \\ 0 & -x \end{pmatrix}, \qquad B_t(x) = \begin{pmatrix} x-t & 0 \\ 0 & -x+t \end{pmatrix}.$$

Then $\|A_t - B_t\|_{C^{0,1}} = \sqrt{2}t$, while the eigenvalue difference at 0 and t changes by the vector $(-2t, 2t)$. Hence

$$[\mathcal{E}(A_t)-\mathcal{E}(B_t)]_\alpha \ge 2\sqrt{2}, t^{1-\alpha}.$$

All these affine curves lie in a common $C^{1,\beta}$ ball. Thus no modulus $o(\delta^{1-\alpha})$ is possible. ☐

The same argument works for normal matrices with the unordered-spectrum metric d_2, and after any fixed Almgren bi-Lipschitz embedding, up to its constants. Sharpness already occurs in the Hermitian subclass.

3 A linear modulus under a uniform spectral gap

Theorem 3 (Uniform-gap $W^{1,q}$ modulus)**.** *Let $U\subset\mathbb{R}^m$ be bounded. Suppose $A, B \in C^1(\overline{U}, \mathrm{Herm}(d))$ satisfy*

$$\|A\|_{C^1}, \|B\|_{C^1} \le C, \qquad \lambda_{i+1}(A(x)) - \lambda_i(A(x)) \ge \gamma, \qquad \lambda_{i+1}(B(x)) - \lambda_i(B(x)) \ge \gamma$$

for all x and i. If $\delta = \|A-B\|_{C^1} \le \gamma/6$, then for $1\le q\le\infty$,

$$\|\mathcal{E}(A)-\mathcal{E}(B)\|_{W^{1,q}(U)} \le C_{d,m,U}\left(1+\frac{C}{\gamma}\right)\delta. \tag{4}$$

Proof. Fix x and i. The circle of radius $\gamma/3$ centered at $\lambda_i(A(x))$ encloses exactly the ith eigenvalue of both $A(x)$ and $B(x)$, by Weyl's inequality. On this circle the two resolvent norms are at most $3/\gamma$ and $6/\gamma$. The Riesz formula and the resolvent identity therefore give the rank-one spectral projections

$$\|P_i(A(x)) - P_i(B(x))\|_{\mathrm{op}} \le \frac{6\delta}{\gamma}. \tag{5}$$

Since all eigenvalues are simple,

$$\partial_j\lambda_i(A) = \mathrm{tr}(P_i(A)\partial_j A), \qquad \partial_j\lambda_i(B) = \mathrm{tr}(P_i(B)\partial_j B).$$

Using $\|P_i(A) - P_i(B)\|_F \le \sqrt{2}\|P_i(A) - P_i(B)\|_{\mathrm{op}}$,

$$\begin{aligned}|\partial_j\lambda_i(A) - \partial_j\lambda_i(B)| &\le \|\partial_j A - \partial_j B\|_F + \|P_i(A) - P_i(B)\|_F\|\partial_j B\|_F \\ &\le \left(1 + \frac{6\sqrt{2}C}{\gamma}\right)\delta.\end{aligned}$$

Together with Hoffman–Wielandt, summing the finitely many indices and integrating proves (4). $\square$

Thus Problem 1.15 has a locally Lipschitz solution on the relatively compact uniform-gap subclasses of K_C^β.

4 A power-sharp Sobolev modulus for diagonal curves

Let I be a bounded interval. Define, for $0 < \varepsilon \le 1$,

$$\Phi_{q,\beta}(\varepsilon) = \begin{cases}\varepsilon^{\beta/(1+\beta)}, & q < 1 + 1/\beta, \\ \varepsilon^{1/q}(1 + |\log\varepsilon|)^{1/q}, & q = 1 + 1/\beta, \\ \varepsilon^{1/q}, & q > 1 + 1/\beta.\end{cases} \tag{6}$$

Lemma 4 (Derivative mass on a thin sublevel set)**.** *For fixed I, M, H, q, β, every $h \in C^{1,\beta}(\overline{I})$ with $\|h'\|_\infty \le M$ and $[h']_\beta \le H$ satisfies*

$$\left(\int_{\{|h|\le\varepsilon\}} |h'|^q\right)^{1/q} \le C_{I,M,H,q,\beta}\Phi_{q,\beta}(\varepsilon). \tag{7}$$

Proof. Fix $s > 0$ and a regular value y. Consecutive solutions of $h(x) = y$ at which $s < |h'(x)| \le 2s$ are separated by at least a constant multiple of $(s/H)^{1/\beta}$: Rolle's theorem supplies a zero of h' between them and the Hölder bound supplies the separation. Hence the number of such roots is at most $1 + C_I H^{1/\beta}s^{-1/\beta}$. The one-dimensional area formula then yields

$$\int_{\{|h|\le\varepsilon,\ s<|h'|\le 2s\}} |h'|^q \le C\varepsilon\big(s^{q-1} + H^{1/\beta}s^{q-1-1/\beta}\big).$$

Split $|h'|$ dyadically above a threshold r; below it, use $|I|r^q$. Set $r = \varepsilon^{\beta/(1+\beta)}$. The second geometric sum is dominated by its bottom, has equal terms, or is dominated by its top according as $q < 1 + 1/\beta$, $q = 1 + 1/\beta$, or $q > 1 + 1/\beta$. This gives (7); the first geometric sum is no larger after adjusting the constant. $\square$

Theorem 5 (Effective diagonal $W^{1,q}$ modulus)**.** *Let $A = \operatorname{diag}(a_1, \dots, a_d)$ and $B = \operatorname{diag}(b_1, \dots, b_d)$ belong to one fixed bounded subset of $C^{1,\beta}(\overline{I}, \operatorname{Herm}(d))$. If $\delta = \|A - B\|_{C^1} \le 1$, then*

$$\|\mathcal{E}(A) - \mathcal{E}(B)\|_{W^{1,q}(I,\mathbb{R}^d)} \le C_{d,I,q,\beta,K}\Phi_{q,\beta}(\delta). \tag{8}$$

The power

$$\theta_{q,\beta} = \min\left\{\frac{\beta}{1+\beta}, \frac{1}{q}\right\} \tag{9}$$

is optimal: no uniform Hölder estimate with exponent larger than $\theta_{q,\beta}$ holds on a fixed bounded $C^{1,\beta}$ ball.

Proof. At almost every x, sorting $a(x)$ and $b(x)$ gives two permutations. Tie sets cause no ambiguity in the weak derivative: if two C^1 scalar functions agree on a measurable set, then their derivatives agree almost everywhere on that set. If their relative permutation inverts indices i, j, then one family orders the pair one way and the other orders it oppositely. Since $\max_k |a_k - b_k| \le \delta$, on the inversion set

$$|b_i - b_j| \le 2\delta.$$

Transform one permutation into the other with at most $d(d-1)/2$ adjacent swaps. The derivative error due to a swap is bounded by $|(b_i - b_j)'|$. Minkowski's inequality and Lemma 4, applied to every pair $b_i - b_j$, give the derivative part of (8). Pairing the same coordinates first adds at most the input derivative error. The zeroth-order term follows from Hoffman–Wielandt and is absorbed into (8).

For the $1/q$ obstruction, use the shifted crossings A_t, B_t from Theorem 2. On $(0,t)$ their ordered eigenvalue derivatives differ by a vector of norm $2\sqrt{2}$, so the $W^{1,q}$ distance is at least $2\sqrt{2}t^{1/q}$ while the input distance is $\sqrt{2}t$.

For the $\beta/(1+\beta)$ obstruction on $I = [0, 2\pi]$, let

$$a_k = k^{-(1+\beta)}, \quad u_k(x) = a_k\sin(kx), \quad v_k(x) = u_k(x) + a_k,$$

and set $A_k = \operatorname{diag}(u_k, -u_k)$, $B_k = \operatorname{diag}(v_k, -v_k)$. These curves stay in a fixed $C^{1,\beta}$ ball, and $\|A_k - B_k\|_{C^1} = \sqrt{2}a_k$. Since $v_k \geq 0$, on the half-periods where $\sin(kx) < 0$ the positive ordered branches have derivative difference $2a_k k \cos(kx)$. Its L^q norm is $c_q k^{-\beta}$, which is a constant multiple of $\delta^{\beta/(1+\beta)}$. The two examples prove (9); at the critical index they leave only the logarithmic factor in (6) undecided. □

5 Verification, limitations, and novelty

The verifier checks the exact power laws in both sharpness families and tests the resolvent-projection and eigenvalue-derivative estimates on random gapped Hermitian matrices. It also compares the shifted-crossing lower bound with (3). All tests pass; they are sanity checks, while the proofs above are analytic.

Adversarial checks. The Hölder result does not use compactness and is dimension-free under the Frobenius/Euclidean convention. The gapped theorem requires a gap for both families (or, equivalently, one family plus a sufficiently small perturbation). The diagonal Sobolev theorem uses one parameter; its proof is based on the one-dimensional area formula and does not automatically extend to multiparameter level sets.

Remaining obstruction. For arbitrary no-gap Hermitian curves, eigenvalue branches need not have a uniform $C^{1,\beta}$ bound even when the matrices do: the eigenvalues of $\left(\begin{smallmatrix} x & \varepsilon \\ \varepsilon & -x \end{smallmatrix}\right)$ have rapidly varying derivatives as $\varepsilon \downarrow 0$. A complete $W^{1,q}$ modulus therefore needs a cluster-stable replacement for the scalar sorting argument. This is the precise unclosed part of Problem 1.15 here.

Bounded novelty audit. The four run indexes and focused primary-source arXiv searches through 13 August 2026 found no later explicit answer to Problem 1.15. The source itself supplies the crossing obstruction for exponents above $1/q$; the optimal $C^{0,\alpha}$ theorem, the gapped estimate, the diagonal upper bound, and the high-frequency $\beta/(1+\beta)$ obstruction were not found in the bounded search. Novelty is plausible, not certified.

Human-review recommendation: send to an expert in matrix perturbation and quantitative Sobolev superposition. The most valuable checks are Lemma 4, the sorting-network reduction, and the scope boundary between diagonal/gapped families and general clusters.

References

[1] A. Parusiński and A. Rainer, *Eigenvalue stability of Hermitian and normal matrices*, arXiv:2603.23056, 2026.

[2] A. J. Hoffman and H. W. Wielandt, *The variation of the spectrum of a normal matrix*, Duke Math. J. **20** (1953), 37–39.